Ashish Jha

B.Tech.(Electronics & Communication)
MSR(Disaster Management & Research)
Signal Processing & Machine Learning
CDMR | IIT Guwahati

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Summary

Currently, a **J**unior **R**esearch **F**ellow under the guidance of Dr. Anirban Dasgupta, Dept. of EEE, IIT Guwahati

- **Research Experience**: Three years in the field of Signal Processing and Machine Learning.
- **Teaching Experience**: Four years in CBSE Board Tutoring

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MSR	Indian Institute of Technology, Guwahati	7.08	2022-2025
B.Tech.(ECE)	NSUT(East Campus), Delhi	7.27 (72.7 %)	2020
Senior Secondary	CBSE, Delhi	65.0 %	2014
Secondary	CBSE, Delhi	8.8 (83.6 %)	2012

Experience

- **UFLEX Pvt. Ltd.** *June – July 2018–19*
  Electronics Department Noida
  – Worked on industrial processes in the electronics division.

- **LearnHill Technologies** *March – May 2024*
  *Student Mentor – **Samagra Shiksha Abhiyan, Assam*** Guwahati
  – Assisted in the implementation of **Tinkering Lab**.
  – Contributed to the RAA project, Govt. of Assam.

- **IIT Guwahati** *July 2022 – May 2025*
  *Research Assistant – **Full-Time*** Guwahati
  – Worked on research projects in electronics and applied sciences.

- **IIT Guwahati** *Aug 2023 – Apr 2024*
  *Maths Faculty – **Super 20 Program (Student Coordinator)*** Guwahati
  – Trained 20 selected students for IIT-JEE 2024 examination.
  – Part of a joint initiative of IIT Guwahati & Kamrup District Administration.

- **IIT Guwahati** *16 – 31 Jan 2025*
  *Student Mentor – **Conducted STEM activities for secondary school students of Assam*** Guwahati
  – Organized hands-on STEM learning activities for high school students.

Projects

- **Crop Damage Assessment of North Bihar Region** *July 2022 – March 2023*
  *Dr. Rishikesh Bharti, **Civil Engg. Dept., IIT Guwahati*** Guwahati
  – Completed

- **Python-based GUI Development for Operational Modal Analysis** *July 2023 – June 2025*
  *Prof. Arunasis Chakraborty, **Civil Engg. Dept., IIT Guwahati*** Guwahati
  – Completed
  – Built an industry-friendly software **SPoTMAn** (***S**ignal **P**rocessing **T**oolbox for **M**odal **A**nalysis*) to perform OMA using various techniques, determine dynamic characteristics of structures, and visualize results.

- **Path Rakshak (Guardian of the Road)** *July 2025 – May 2026*
  *Dr. Anirban Dasgupta, **Assistant Professor, EEE Dept., IIT Guwahati*** Guwahati
  – **Ongoing**
  – Building a Machile Learning model for enhancing public transport safety with real-time ambiance and driver monitoring.

- **Early Lung Cancer Detection & Diagnosis** *February 2026 – Present*
  *Dr. Kaushik J. Kataki, **Radiation Oncologist, Dr. B. Borooah Cancer Institute*** Guwahati
  – **Ongoing**
  – Building a Machine Learning model for early detection of Lung Cancer

Technical Skills

- **Programming**: Python | MATLAB | SQL | CLIPS*
- **Data Analytics / Visualisation**: MS Excel | MS Word | R
- **Statistical Software**: SPSS | STATA
- **GIS Tools**: ENVI | ArcGIS
- **Machine Learning**: Python (NumPy | Pandas | TensorFlow | PyTorch)

** Elementary proficiency*

Certification Courses

- **Mathematics**: Linear Algebra | Advanced Calculus | Discrete Maths | Probability & Random Processes
- **Electronics**: Signal Processing (MATLAB) | App Building using MATLAB
- **Computer Science**: Programming with Python | Data Analytics (Google) | Excel Essentials for Business
- **Disaster Management**: Advanced Remote Sensing | Applications of AI in Disaster Management | Social and Policy Aspects of Disaster Management

Positions of Responsibility

- **Core Team Member**, Prakriti – "Techno-Environmental Club" (IIT Guwahati) *Apr. 2023 – June 2024*
- **Team Member**, TEDx'24 Content Team (IIT Guwahati) *Aug. 2023 – June 2024*

Extra Curriculars

- **Deep Learning for Vision & Language Modelling**, Winter School @ CSE Dept. (IIT Guwahati) *6th–13th Jan. 2025*
- **IEEE Sensors Council Workshop**, Emerging Nano/Micro Sensors and Devices (IWS-2023) *December 2023*
- **Research and Industrial Conclave – Integration'2023**, Volunteer *May 2023*

Academic Achievements

- **GATE(IN)**, AIR-935 *2022*
- **GATE(EC)**, AIR-1431 *2026*
- **JEE Mains**, Qualified *2016*
- **UGC-NET**, Electronic Science *2024*